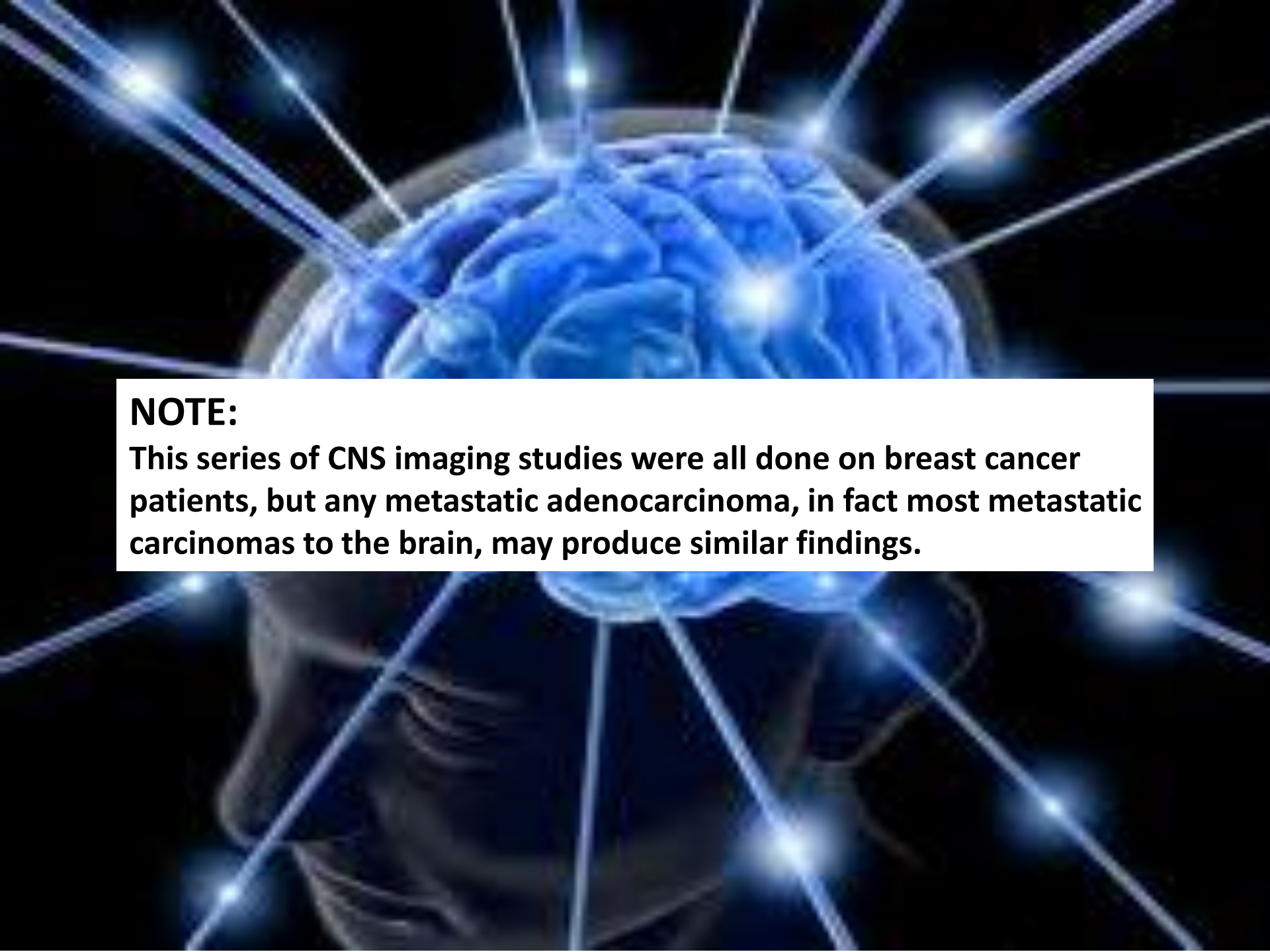




BRAIN METASTASES

IMAGING FINDINGS

LIISA A. RUSSELL, MD

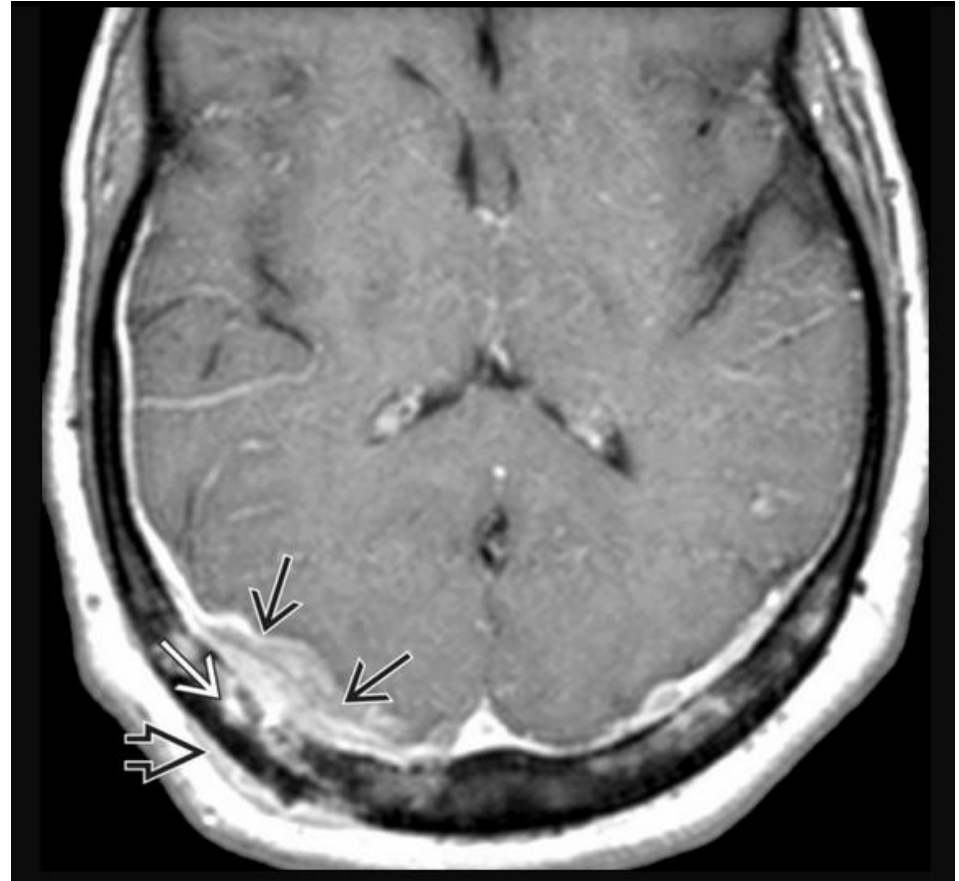
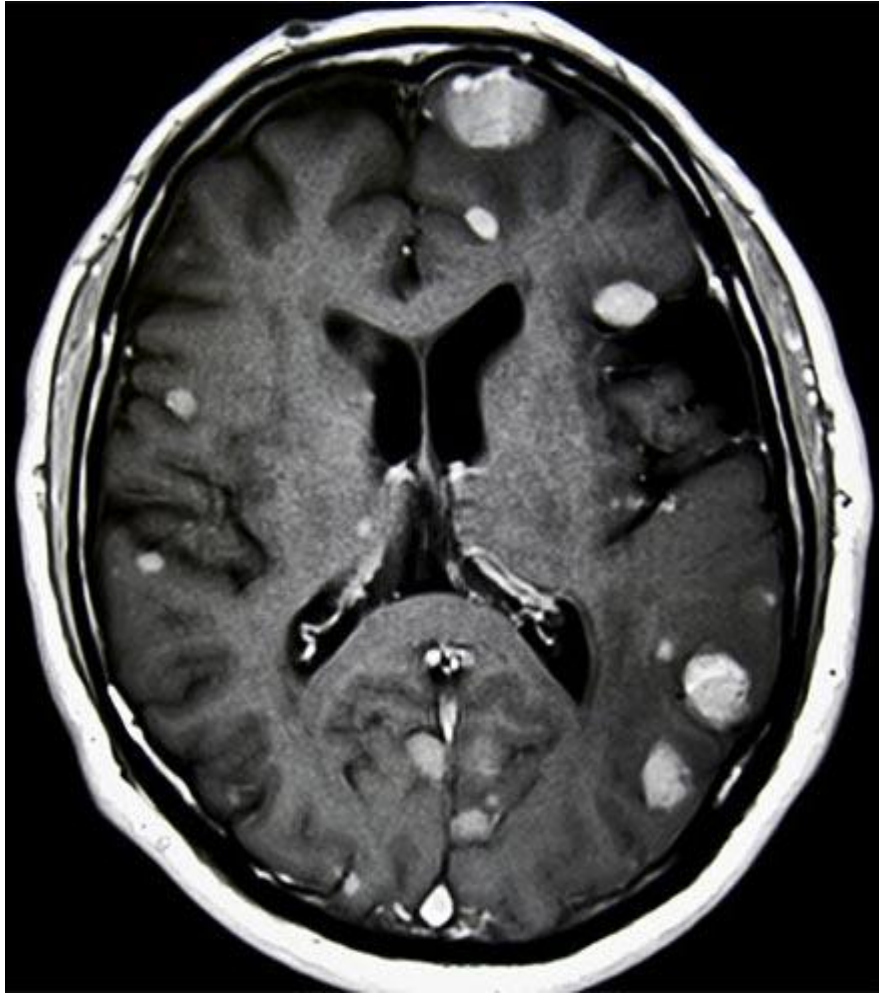


NOTE:

This series of CNS imaging studies were all done on breast cancer patients, but any metastatic adenocarcinoma, in fact most metastatic carcinomas to the brain, may produce similar findings.

MULTIFOCAL PARENCHYMAL AND MENINGEAL LESIONS

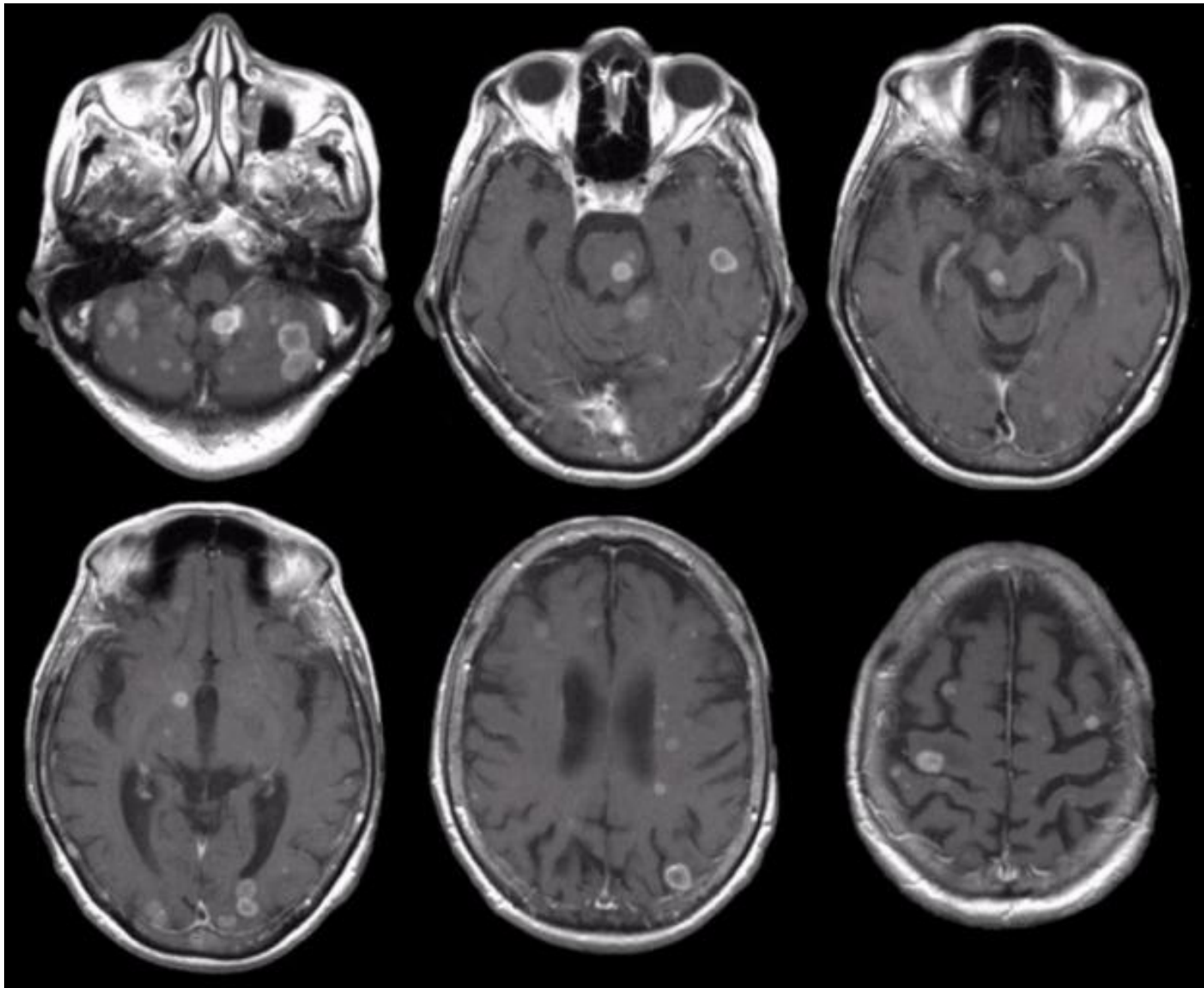
MRI T1 WITH CONTRAST



Diff. Dx: metastatic tumor vs. inflammatory/infectious lesion.
This patient turned out to have metastatic breast cancer to the brain.

BREAST CANCER METASTASES TO BRAIN

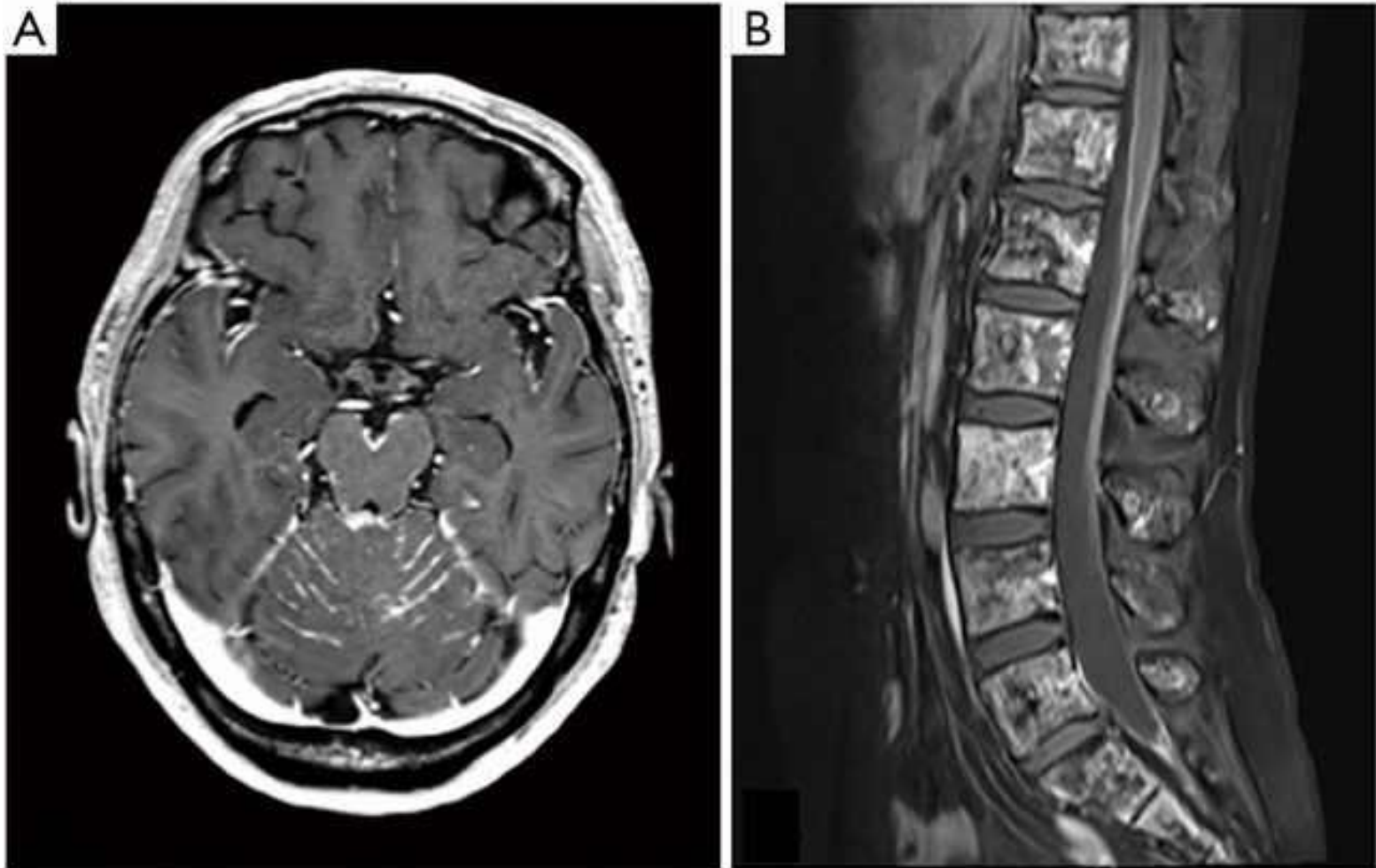
T1-WEIGHTED AXIAL MRI WITH GADOLINIUM CONTRAST



Numerous enhancing lesions in the cortex, deep white mater, brainstem and cerebellum.

LEPTOMENINGEAL CARCINOMATOSIS IN BREAST CANCER

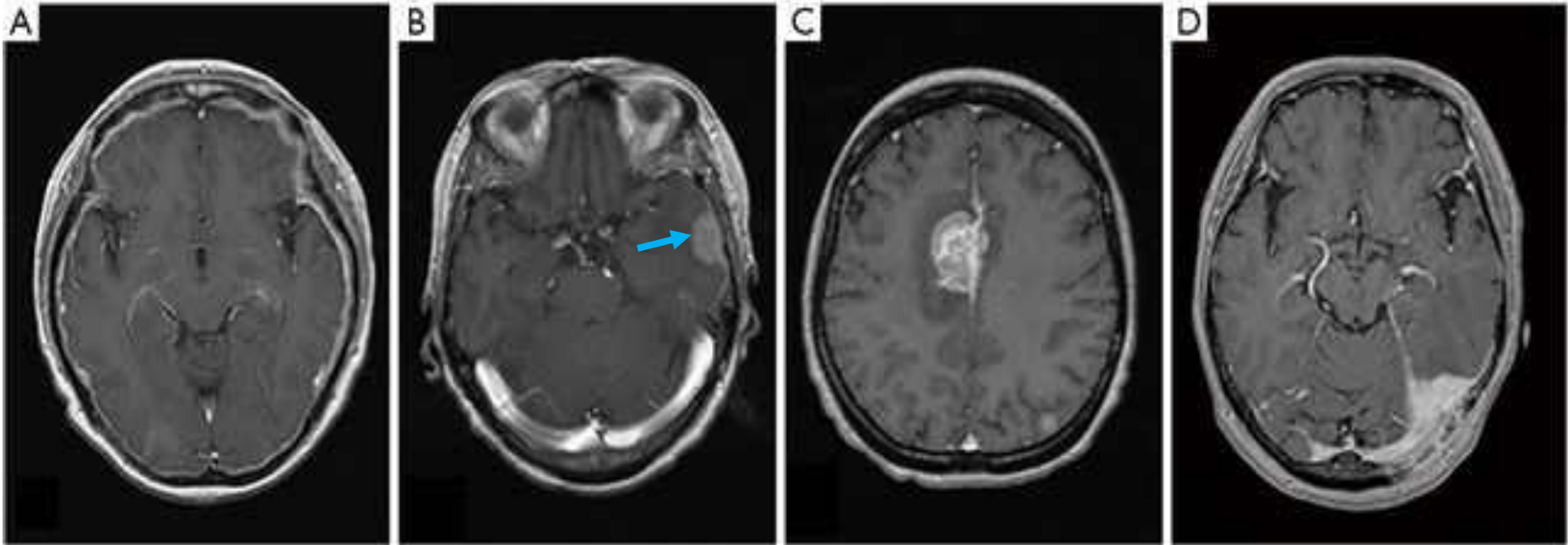
T1-WEIGHED AXIAL MRI WITH GADOLINIUM CONTRAST



- A. Linear enhancement on the surface of midbrain and cerebellar folia
- B. Linear enhancement, so-called railway sign, on the surface of the spinal cord

INTRACRANIAL DURAL METASTASIS OF BREAST CANCER

T1-WEIGHTED MRI WITH GADOLINIUM CONTRAST



- A. Axial enhanced MRI shows a typical diffuse linear dural enhancement
- B. MRI demonstrates a tumor and dural tail sign with brain edema in the left middle fossa convexity. Compare this to the imaging findings in Review Case 2.
- C. MRI shows a tumor attached to the right frontal falx cerebri.
- D. MRI demonstrates partial dural thickening in the cerebellar tentorium.